

SetuCredit

High-Level Design (HLD) Document

System Architecture & Implementation Specification for Rural Credit Inclusion

DOCUMENT

High-Level Design

TRACK

Financial Inclusion & DPI

OWNER

SetuCredit Core Team

STATUS

Hackathon Submission

1 Purpose & Business Scope

SetuCredit is a middleware orchestration layer connecting unbanked and underserved micro-borrowers in India with formal banking institutions. By leveraging India's Digital Public Infrastructure (DPI), specifically the RBI Unified Lending Interface (ULI), Sahamati Account Aggregator (AA), and Bhashini AI, the platform eliminates manual paper-based verification and computes risk scores directly from non-financial operational records.

RBI ULI

Authoritative, standardized access to lending data and lender connectivity.

Account Aggregator

Revocable, consent-driven data sharing via Sahamati-registered rails.

Bhashini AI

Voice and language layer so borrowers can transact in their own dialect.

2 System Architecture

Requests flow top to bottom; the orchestrator fans out to DPI sources in parallel.

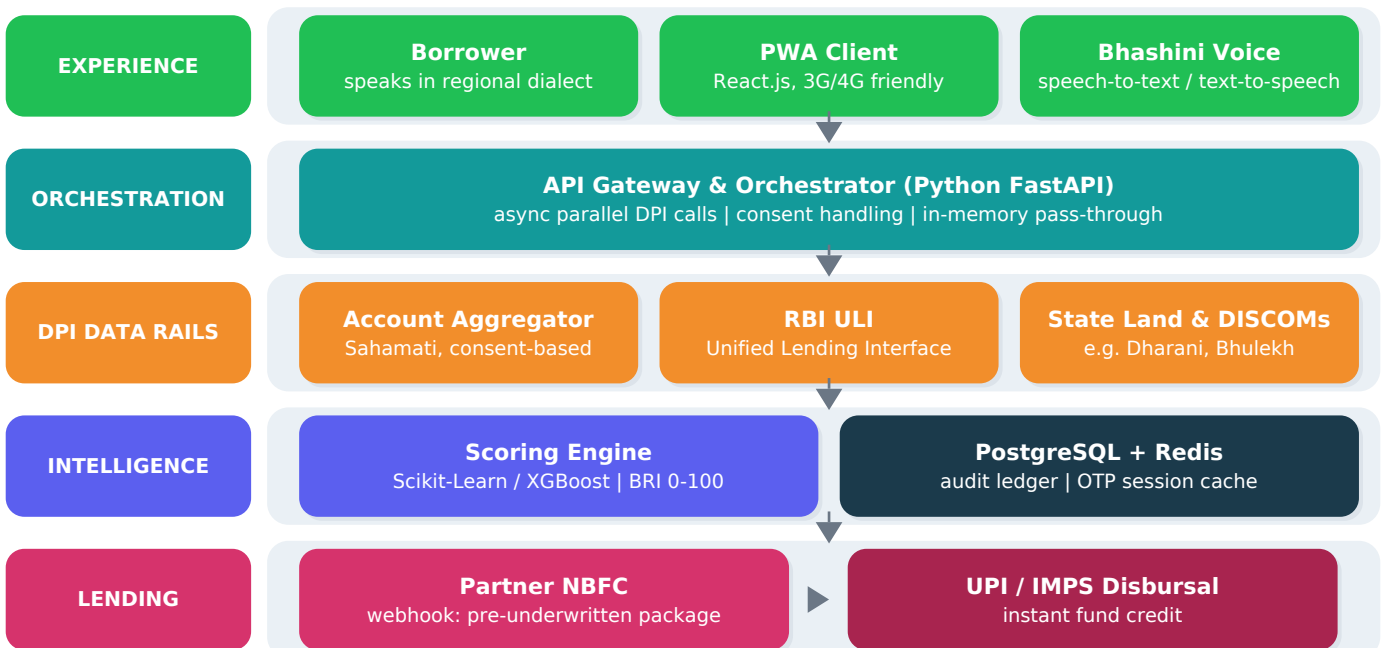


Figure 1: SetuCredit logical architecture (BRI = Borrower Readiness Index).

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End-to-End Request Flow

Six stages take a borrower from first utterance to credited funds, with consent captured before any data is pulled.

#	Stage	What happens	Component
1	Voice onboarding	Borrower opens the Progressive Web App (PWA) and speaks in a regional dialect via the Bhashini voice engine.	PWA + Bhashini
2	Identity	Authenticates identity via Aadhaar OTP (DigiLocker verification).	DigiLocker / Aadhaar
3	Consent	Grants revocable consent via Account Aggregator to pull utility, cooperative, and land data.	Account Aggregator
4	Data retrieval	SetuCredit backend coordinates simultaneous queries to state land registries (e.g., Dharani) and utility boards.	FastAPI orchestrator
5	Scoring	Scoring microservice evaluates input vectors and computes a Borrower Readiness Index (0-100).	Scoring engine
6	Disbursal	Pre-underwritten package triggers the partner NBFC webhook; funds disburse instantly via UPI/IMPS.	NBFC + UPI/IMPS

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Core Technology Stack

Component	Technology	Role & Justification
Client Frontend	React.js / PWA	Low-storage, runs smoothly on 3G/4G networks
Vernacular Voice	Bhashini API	Speech-to-text / text-to-speech in 12+ Indian languages
API Gateway	Python (FastAPI)	Async architecture handles parallel DPI endpoint requests
Data Rails	RBI ULI + AA	Authoritative access to state land registries and DISCOMs
Scoring Engine	Scikit-Learn / XGBoost	Explainable machine learning model for risk indexing
Ledger & Cache	PostgreSQL + Redis	Postgres for audit ledgers; Redis for fast OTP session caching

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Security & Compliance

Data Minimization SetuCredit operates strictly in memory as a pass-through orchestration engine. Borrower non-financial data is not persistently stored on application servers.	DPDP Act Compliance Data pulls require explicit, time-bound, and revocable consent from the borrower.	Encryption Enforces TLS 1.3 in transit and AES-256 for all stored audit logs and ledgers.
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Design principle: consent first, pass-through by default, explainable scoring, instant outcomes for the borrower.